Example 3:

Record the observations in the form of levelling field book and obtain reduced level of each point. Apply usual check.

Reading of inverted staff on a point 'A' whose reduced level 15.995 m is 3.220. Reading of staff on a point 'B' on ground is 0.805. Change the instrument and the reading of staff on point 'B' is again 1.280. Inverted Reading of staff on point C is 3.695.

Type I: Height of Collimation method

Solution:

The observation are tabulated and reduced levels are calculated as follows.

Station	BS	IS	FS	HC	RL	Remarks
A	–3.220			12.775	15.995	BM – 1
B	1.280		0.805	13.250	11.970	CP
C			–3.695		16.945	BM – 2
Total	–1.940		–2.890			

(Negative sign indicates inverted staff reading)

Calculations

HC for I set up = RL of A + BS of A = 15.995 + (-3.220) = 12.775

RL of B = HC for I set up – FS of B = 12.775 - 0.805 = 11.970

HC for II set up = RL of B + BS of B = 11.970 - 1.280 = 13.250

RL of C = HC for II set up – FS of C = 13.250 – (-3.695) = 16.945

Arithmetic check

Σ BS ~ Σ FS = first RL ~ last RL

1.940 ~ 2.890 = 16.945 ~ 15.995

0.950 = 0.950

Hence checked

Type II Rise and fall method

Station	BS	IS	FS	Rise (+)	Fall (–)	RL	RL
A	–3.220					15.995	BM – 1
B	1.280		0.805		4.025	11.970	CP
C			–3.695	4.975		16.945	BM – 2
Total	–1.940		–2.890	4.975	4.025		

(Negative sign indicates inverted staff reading)

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